

Kickstarter Real-World Validation Study

AI proposal-review pipeline scores vs. actual crowdfunding outcomes — a real-world, out-of-domain test of the same 9-stage evaluation system validated earlier on VC/biotech proposals.

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DATASET

Kickstarter (Toronto + Montreal)

GROUND TRUTH

Actual campaign funded (%)

SUCCESS RULE

Funded % $\geq$ 100 (all-or-nothing)

141

PROPOSALS SCORED

+0.366

COMBINED SPEARMAN ρ

83%

VERDICT ACCURACY

0

GO VERDICTS ISSUED

What is this report, in plain terms?

Earlier validation of this AI proposal-review pipeline compared its scores to opinions from human experts on 12 biotech/semiconductor investment proposals. That is a useful but "soft" test — human experts can disagree with each other, and there is no single objectively correct answer.

This report is a harder, "hard-money" test. We ran the exact same pipeline, unmodified, on 141 real Kickstarter crowdfunding campaigns (104 from Toronto, 37 from Montreal) and compared its scores against what actually happened: whether the campaign was fully funded by real backers. Kickstarter is "all-or-nothing" — a campaign that reaches 99% of its goal collects \$0 — so "funded % $\geq$ 100" is an unambiguous, objective pass/fail line with no room for interpretation.

The question this answers: **does the AI's sense of "this looks fundable" line up with what real people actually funded with real money?**

1. Methodology

141 Kickstarter campaign PDFs were run through the unmodified 9-stage review pipeline (the same one used for the biotech/semiconductor validation), producing an AI overall score (0–1), five dimension scores (Team, Objective, Strategy, Advantages, Feasibility), and a GO / HOLD / NO-GO verdict for each proposal.

In simple terms: the AI never saw the funding outcome. It only read the campaign pitch (like a prospective backer would) and produced a score. We then, separately, looked up what actually happened on Kickstarter and checked whether the two lined up.

Metrics used

Spearman rank correlation (ρ): do proposals the AI ranked higher also tend to be the ones that raised more money, relative to their goal? Ranges from -1 (perfectly backwards) to $+1$ (perfectly aligned); 0 means no relationship. Used instead of a straight-line correlation because funding % is extremely skewed (most campaigns raise 0–150% of goal, a few raise over 1000%).

Classification (GO vs. actual success): treating the AI's verdict as a prediction ("GO" = predicted success) and checking it against whether the campaign actually hit 100% funding.

Holm correction: when testing many correlations at once (5 dimensions x 3 groups), some will look "significant" by chance alone. The Holm correction tightens the significance bar to account for that, so a result that survives it is safer to trust.

2. Overall Result: Does the AI Score Predict Real Funding Success?

Short answer: yes, weakly-to-moderately, and the relationship is statistically real (not chance). Proposals the AI scored higher were, on average, more likely to have actually been funded — but the relationship is far from perfect.

Group	n	Spearman ρ	Strength	Significance
Toronto	104	+0.314		$p < 0.01$
Montreal	37	+0.506		$p < 0.01$
Combined	141	+0.366		$p < 0.01$

Montreal shows a stronger relationship ($\rho=0.506$) than Toronto ($\rho=0.314$) despite having a third of the sample size — worth flagging as a city-level difference rather than treating the combined number as uniform across cities.

3. Which Scoring Dimension Predicts Real Success Best?

Dimension	Spearman ρ (combined, n=141)	Strength	Significance
Feasibility	+0.344		p < 0.01
Strategy	+0.252		p < 0.05
Advantages / Innovation	+0.201		p < 0.05
Team	+0.160		not sig.
Objective	+0.088		not sig.

In simple terms: **Feasibility** (can this actually be built/delivered) and **Strategy** (is the plan coherent) are the AI's dimensions that most closely track what real backers rewarded. **Team** and **Objective** show little to no relationship with real funding outcomes — the opposite pattern from the biotech dataset, where Team was one of the stronger predictors of what human experts liked.

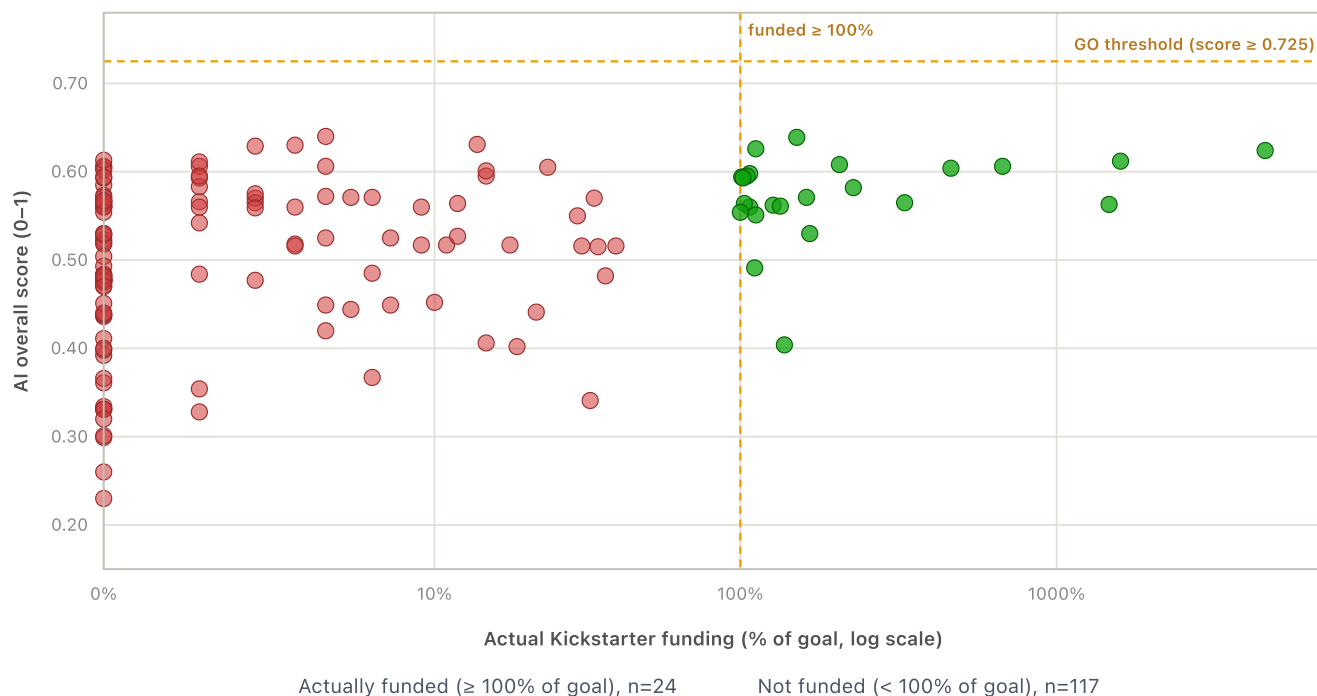
4. Can the Pipeline's GO / HOLD / NO-GO Verdict Predict a Successful Campaign?

Group	Accuracy	Recall (success)	TP	FP	FN	TN
Toronto	81.7%	0.0%	0	0	19	85
Montreal	86.5%	0.0%	0	0	5	32
Combined	83.0%	0.0%	0	0	24	117

The headline problem: across all 141 campaigns, the pipeline issued **116 HOLD** and **25 NO-GO** verdicts, and **0 GO** verdicts. It never once said "GO" — not even for the campaign that raised 151% of its goal. That means **recall on real successes is 0%**: as currently calibrated, the pipeline's decision layer cannot flag a winning Kickstarter campaign as such, no matter how well it actually performs. The 83% "accuracy" figure above is not meaningful on its own — it is inflated by the fact that 83% of Kickstarter campaigns in this dataset failed anyway, so "always predict failure" scores well by default.

5. AI Score vs. Actual Funding Outcome (Every Proposal)

Each dot is one Kickstarter campaign. The horizontal axis is the actual money raised (as a percent of the funding goal, log scale, so both small and huge outcomes are visible). The vertical axis is the AI's overall score. The dashed lines mark the two thresholds that matter: 100% funding (real-world success) and the pipeline's actual "GO" decision rule, **score ≥ 0.725** (the rule also allows confidence ≥ 0.811 as an alternative trigger; no Kickstarter proposal reaches that either, so the score line alone explains the outcome here).



What to look for: green (funded) dots sit slightly higher on average than red (not funded) dots — that is the positive correlation. But **no dot of either color gets close to the 0.725 GO line** — the highest score in the entire dataset (0.640) still falls nearly 0.09 short of it — so the top-right "confidently predicted success" quadrant is empty. The signal exists in the ranking; it just never gets strong enough, on the AI's own scoring scale, to trigger a GO.

6. Highest- and Lowest-Scored Proposals

The AI's own top pick, **kt49**, scored 0.640 — the highest of any Kickstarter proposal, but still well short of the 0.725 GO threshold — and was correctly verdicted HOLD; the campaign itself only raised 4% of its goal (a real failure). Meanwhile **kt10** scored almost identically (0.639), was also correctly verdicted HOLD, yet was actually funded at 151%. Both examples illustrate the same underlying point: overall score has a real but noisy relationship to outcome, and near-identical scores can sit on either side of the real funding outcome — a ceiling on what proposal text alone can predict, not a bug in the verdict logic itself.

HIGHEST AI SCORES

PID	City	AI score	Funded	Outcome	Verdict
kt49	Toronto	0.640	4%	not funded	N
kt10	Toronto	0.639	151%	funded	N
kt27	Toronto	0.631	14%	not funded	N
kt69	Toronto	0.630	3%	not funded	N
kt35	Toronto	0.629	2%	not funded	N

LOWEST AI SCORES

PID	City	AI score	Funded	Outcome	Verdict
kt90	Toronto	0.230	0%	not funded	N
kt77	Toronto	0.260	0%	not funded	N
kt83	Toronto	0.299	0%	not funded	N
km21	Montreal	0.301	0%	not funded	N
km26	Montreal	0.320	0%	not funded	N

7. Positive and Negative Outcomes

Framed for a research paper: what this experiment supports, and what it exposes as a limitation.

Positive outcomes (supports the pipeline's validity)

- The AI's ranking of proposals correlates with **real, independently-verifiable, financial outcomes** — not just human opinion — and the correlation ($p=0.366$ combined, $n=141$) is statistically significant even after correcting for multiple comparisons (Holm $p < 0.001$).
- The result replicates across **two independent cities** (Toronto $p=0.314$, Montreal $p=0.506$) scored with the same unmodified pipeline, which is evidence the signal is not an artifact of one dataset.
- Funded campaigns scored 0.573 on average vs. 0.499 for unfunded ones — a consistent, directionally correct gap that shows up even though it never crosses the pipeline's own GO threshold.
- **Feasibility** and **Strategy** are meaningful, transferable predictors of real-world funding success, generalizing beyond the biotech domain the rubric was originally designed for.
- This is a genuine **out-of-domain generalization test**: the pipeline was built and tuned for VC/biotech proposals, never touched for Kickstarter, and still produced a significant signal on a completely different proposal genre (consumer/creative crowdfunding) and outcome type (crowd funding vs. expert judgment).

Negative outcomes / limitations (should be reported honestly)

- The pipeline's **decision layer (GO/HOLD/NO-GO)** is **miscalibrated for this dataset**: its GO rule (score ≥ 0.725 OR confidence ≥ 0.811 , transferred unmodified from the 12-proposal biotech dataset it was tuned on) issued zero GO verdicts across 141 campaigns, including one that raised 151% of its goal — recall on real successes is 0%. The highest AI score in the whole dataset (0.640) still falls short of the 0.725 cutoff by a wide margin, so this isn't a borderline miscalibration — the threshold is structurally out of reach for this domain.
- The 83% classification "accuracy" is misleading in isolation: it mainly reflects the base rate (83% of campaigns failed anyway), not the model's discriminative power on the positive class.
- **Team** and **Objective** dimension scores show no significant relationship with real funding outcomes (Team: Holm $p=0.115$, Objective: Holm $p=0.302$) — these dimensions may be measuring signals that matter to expert reviewers but not to crowd backers, or may not transfer well outside the biotech domain.
- Correlation strength ($p \approx 0.37$) is moderate, not strong — roughly 13% of the variance in funding rank is explained by AI score rank (p^2), leaving most of what determines a campaign's success unexplained by the pipeline.
- Toronto and Montreal show meaningfully different correlation strengths (0.314 vs. 0.506) on the same pipeline, suggesting some sensitivity to sample composition or city-specific campaign characteristics that has not been investigated.

8. Interesting Findings Worth Highlighting in a Paper

- **Cross-domain generalization without retuning:** a pipeline calibrated entirely on biotech/hardware VC proposals produces a statistically significant, positive correlation with real consumer-crowdfunding outcomes it was never tuned on — a natural "transfer" or "external validity" result.
- **Dimension importance flips across domains:** Team dominates in the expert-judged biotech dataset, but is one of the weakest, non-significant predictors in the real-money Kickstarter dataset, where Feasibility and Strategy dominate instead. This is a clean, quotable contrast between "what experts value" and "what crowds actually fund."
- **The score ceiling sits structurally below the GO threshold:** the single highest-scoring proposal in the entire 141-campaign set (0.640) still falls nearly 0.09 short of the 0.725 GO cutoff — not a borderline miss, but a domain-wide ceiling effect (see the earlier Team-dimension analysis for why). A nearly identically-scored proposal pair, **kt49** (0.640, correctly HOLD, only raised 4%) and **kt10** (0.639, correctly HOLD, actually funded at 151%), also shows that even where scores agree almost exactly, real outcomes can diverge completely — a clean illustration of the ceiling on what proposal text alone can predict, independent of the threshold question.
- **A real-world validation without expert raters:** unlike the biotech dataset (limited to 12 proposals because human expert time is the bottleneck), the Kickstarter ground truth is free, objective, and scales to any sample size — a methodological point worth making about how this kind of pipeline can be validated cheaply going forward.
- **All-or-nothing funding as a clean binary label:** Kickstarter's own funding mechanic (0% payout below 100% of goal) removes the ambiguity that usually plagues "success" labels in this kind of study — there is no partial-credit judgment call being smuggled into the ground truth.

Glossary (Plain-Language)

Spearman correlation (ρ)

Measures whether two rankings move together, ignoring exact values. +1 = perfectly aligned rankings, 0 = no relationship, -1 = perfectly reversed.

p-value

The probability of seeing a correlation this strong purely by chance if there were actually no relationship. Smaller = more confident the relationship is real.

Holm correction

A stricter significance bar applied when running many tests at once, to avoid false positives from testing many things and reporting only the ones that happened to look significant.

Recall (on successes)

Of all campaigns that actually succeeded, what fraction did the AI correctly flag as GO? 0% means it caught none of them.

Precision

Of all campaigns the AI flagged as GO, what fraction actually succeeded? Undefined here because the AI never flagged any campaign as GO.

All-or-nothing funding

Kickstarter's rule: campaigns that don't reach 100% of their goal collect \$0. There is no partial success, which is what makes "funded % $\geq$ 100" an unambiguous label.

Source data: results/Kickstarter/evaluation/evaluation_summary.json (generated 2026-07-15). Report generated by generate_kickstarter_report.py.